PROJECT REPORT

ON

Reunite: Smart surveillance for Missing Person Tracking

SUBMITTED TO

THE SAVITRIBAI PHULE PUNE UNIVERSITY, IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING
IN
INFORMATION TECHNOLOGY

BY

SASHIKANT KAILASH MORE (B400310814)

ARADHYA SANTOSH SONAR (B400310748)

CHETAN KAILAS WANI (B400310873)

UNDER THE GUIDANCE OF

MRS. ASHWINI BHAMRE

DEPARTMENT OF INFORMATION TECHNOLOGY
Modern College of Engineering, Shivajinagar

A.Y. 2025-26

P.E.S Modern College of Engineering, Pune-05
Department of Information Technology
A.Y 2025-26

CERTIFICATE

This is to certify that the Project Entitled

"Reunite: Smart surveillance for Missing Person Tracking"

Submitted by

Sashikant Kailash More	(B400310814)
Aradhya Santosh Sonar	(B400310748)
Chetan Kailas Wani	(B400310873)

is a Bonafide work carried out by students under the supervision of **Mrs. Ashwini Bhamre** and it is submitted towards the partial fulfillment of the requirement of Bachelor of Engineering (Information Technology) Project.

This project report has not been earlier submitted to any other Institute or University for the award of any degree or diploma.

Mrs. Ashwini Bhamre
Internal Guide

External Examiner

Dr. Mrs. S. S. Bhavsar
HOD-IT & AIML

Prof. Dr. K. R. Joshi
Principal

Place: Pune
Date:

ACKNOWLEDGMENT

We are delighted to submit our preliminary project report on "**Reunite: Smart surveillance for Missing Person Tracking**". We would like to express our sincere gratitude to our internal guide, **Mrs. Ashwini Bhamre**, for her constant guidance, valuable suggestions, and encouragement throughout the preparation of this report. Her support and timely advice have been instrumental in shaping our work.

We are also profoundly thankful to **Dr. S. S. Bhavsar**, Head of the Information Technology Department, Modern College of Engineering, for her indispensable support and constructive suggestions that have greatly con-tributed to the success of this project.

Our special thanks are due to all the college staff for providing essential resources such as a well-equipped laboratory, necessary software platforms, and continuous internet connection, which facilitated the successful completion of our project report.

Aradhya Santosh Sonar (B400310748)
Shashikant Kailash More (B400310814)
Chetan Kailas Wani (B400310873)

ABSTRACT

Reunite: Smart Surveillance for Missing Person Tracking is an AI-based surveillance system developed to improve the process of identifying and locating missing individuals. Traditional investigation methods such as poster distribution, public announcements, and manual CCTV monitoring are often time-consuming and less efficient. The proposed system aims to overcome these limitations by integrating intelligent facial recognition with real-time surveillance monitoring.

The system is implemented as a web-based platform using Django and SQLite, along with deep learning–based face detection and recognition models. When a missing person complaint is registered, the guardian's details and multiple photographs of the missing individual are securely stored in a centralized database. These stored facial records are then continuously compared with faces detected from live CCTV video streams.

The surveillance module uses RetinaFace for face detection and ArcFace for facial feature extraction and recognition. Whenever a potential match is identified, the system automatically generates an alert containing the detected image, location details, timestamp, and case information. The alert is then forwarded to the respective police station for immediate action.

The proposed system reduces manual effort involved in missing person investigations and improves the speed and reliability of identification. By combining deep learning, real-time monitoring, and automated alert generation, the system supports law enforcement agencies in conducting faster and more effective search operations, thereby increasing the chances of reuniting missing individuals with their families.

Keywords: Facial Recognition, Deep Learning, RetinaFace, ArcFace, Smart Surveillance, Missing Person Tracking, Django, Real-Time Monitoring, Web Application.

CONTENTS

LIST OF ABBREVIATIONS

Sr. No.	Abbreviation	Full Form
1	IT	Information Technology
2	AI	Artificial Intelligence
3	DFD	Data Flow Diagram
4	UML	Unified Modeling Language
5	ERD	Entity Relationship Diagram
6	SRS	Software Requirement Specification
7	GUI	Graphical User Interface
8	API	Application Programming Interface
9	DBMS	Database Management System
10	CNN	Convolutional Neural Network
11	CCTV	Closed-Circuit Television
12	SQL	Structured Query Language
13	SMTP	Simple Mail Transfer Protocol
14	AWS	Amazon Web Services
15	HTML	HyperText Markup Language
16	CSS	Cascading Style Sheets
17	JS	JavaScript
18	OTP	One-Time Password
19	CNN	Convolutional Neural Network
20	MTCNN	Multi-Task Cascaded Convolutional Network
21	LBPH	Local Binary Pattern Histogram
22	SVM	Support Vector Machine
23	FPS	Frames Per Second
24	GPS	Global Positioning System
25	CRUD	Create, Read, Update and Delete
26	SSD	Solid State Drive
27	RAM	Random Access Memory
28	FPN	Feature Pyramid Network
29	RetinaFace	Single-Stage Face Detection Framework
30	ArcFace	Additive Angular Margin Loss Face Recognition Model

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1

Introduction

1.1 Motivation

Every Every year, many people, including children, elderly individuals, and mentally challenged persons, are reported missing. Such incidents not only create emotional distress for families but also increase the workload on law enforcement agencies. Conventional methods used for locating missing individuals, such as poster circulation, newspaper advertisements, public announcements, and manual monitoring of CCTV footage, are often slow, inefficient, and highly dependent on human effort. In many cases, delays in identification reduce the chances of locating missing persons quickly.

The proposed system integrates real-time surveillance monitoring with AI-based face detection and recognition techniques. By continuously analyzing CCTV video streams and comparing detected faces with stored missing person records, the system can identify potential matches and automatically generate alerts. This reduces manual investigation efforts, improves response time, and increases the overall efficiency of case handling.

The project aims to transform traditional missing person investigations into a more proactive, technology-driven, and intelligent process, thereby helping law enforcement agencies reunite missing individuals with their families more effectively and within less time.

.

1.2 Objectives

The existing missing person tracking process mainly relies on manual investigation methods, printed posters, and public announcements, which are time-consuming and often unreliable in large-scale surveillance environments. The lack of a centralized digital system and automated monitoring mechanisms further delays identification and reduces investigation efficiency.

To address these challenges, the proposed system, “Reunite: Smart Surveillance for Missing Person Tracking,” is designed to automate and simplify the identification process using AI-based facial recognition and centralized case management. The integration of surveillance cameras with intelligent detection models enables continuous monitoring, faster identification, and automated alert generation.

The main objectives of the proposed system are as follows:

1. To design and develop a centralized web-based platform for storing and managing missing person records securely.
2. To provide access to authorized police personnel for case registration, monitoring, and management.
3. To implement deep learning–based facial recognition models for accurate face detection and facial feature extraction.
4. To integrate CCTV surveillance systems for continuous real-time face monitoring and comparison.
5. To generate automated alerts containing the detected person's image, location, timestamp, and case details whenever a match is identified.
6. To maintain data security, privacy, and controlled access to sensitive information stored in the system.
7. To reduce manual investigation efforts, dependency on poster-based searches, and delays in locating missing individuals.
8. To improve the speed, accuracy, and reliability of missing person identification through intelligent surveillance technologies.

CHAPTER 2

Literature Survey

2.1 Related Work

Advancements in face detection and recognition technologies have significantly improved the performance of intelligent systems used for missing person identification and surveillance applications. Researchers have explored both traditional machine learning methods and modern deep learning–based approaches to improve detection accuracy, recognition reliability, and real-time processing capabilities.

Earlier face detection and recognition systems commonly used techniques such as Haar Cascade classifiers, Principal Component Analysis (PCA), and Local Binary Pattern Histogram (LBPH) because of their low computational requirements and simple implementation. Bharathwaj et al. [6] developed a missing child identification system using Haar Cascade for face detection and LBPH for face recognition. Although these approaches performed effectively in controlled environments, their recognition accuracy decreased under variations in lighting conditions, facial pose, and partial occlusion.

The introduction of Convolutional Neural Networks (CNNs) brought major improvements in facial recognition systems by enabling automatic feature extraction and learning of complex facial patterns. Multistage face detection frameworks such as MTCNN improved face localization and alignment accuracy under different environmental conditions. However, the sequential processing structure of multistage detectors increased computational complexity and limited their suitability for large-scale real-time applications [4].

To overcome these limitations, single-stage face detection frameworks such as RetinaFace were introduced. RetinaFace uses Feature Pyramid Networks (FPN) to detect faces at multiple scales while simultaneously predicting facial landmarks for alignment. Research studies indicate that RetinaFace achieves high detection accuracy even for low-resolution and partially occluded faces, making it suitable for real-time surveillance applications [1], [2].

In the area of face recognition, embedding-based deep learning models have shown better performance compared to traditional classification techniques. ArcFace introduced an additive angular margin loss function that improves inter-class separability and intra-class compactness within the embedding space [1]. Studies show that ArcFace-based embeddings provide reliable recognition performance under challenging real-world conditions such as pose variation, illumination changes, and low-quality surveillance footage [1], [3].

Similarity-based matching techniques have also become an important part of modern recognition systems. Cosine similarity is widely adopted for embedding comparison due to its computational efficiency and scale-invariant properties. Several existing systems use cosine similarity with normalized facial embeddings to support open-set recognition in large-scale surveillance environments [3], [5].

Researchers have also explored distributed and real-time processing architectures to improve system scalability and operational efficiency. Kazanskiy et al. [12] analyzed distributed stream-processing frameworks for real-time face detection, while Kanagamalliga et al. [15] proposed edge-assisted surveillance architectures combining local face detection with cloud-based recognition to balance computational performance and latency. Optimized CNN-based recognition pipelines have further demonstrated improved real-time processing capabilities with high recognition accuracy [14].

Despite these advancements, existing systems still face challenges related to computational efficiency, threshold selection, scalability, and reliable recognition under unconstrained surveillance conditions. These limitations highlight the need for an intelligent, scalable, and real-time surveillance framework capable of performing accurate missing person identification with reduced manual intervention.

Table 2.1: Literature Survey Table

Year	Paper Name	Methodology	Advantages	Limitations
2025	Face Recognition in Online Soccer Streaming for Piracy Detection [11]	RetinaFace + Haar Cascade with FaceNet embeddings and cosine similarity	High precision, real-time recognition (~20 FPS), low false positives	Reduced recall due to strict threshold calibration
2025	Multiscale Facial Detection Using RetinaFace Architecture [2]	RetinaFace with Feature Pyramid Networks (FPN) and facial landmark alignment	High accuracy for low-resolution and occluded faces	Requires GPU resources for real-time deployment
2025	AI Finder: An Intelligent Framework for Finding Missing Persons Using Face Match Algorithm [3]	AI-based face matching with embedding comparison	Improved identification accuracy and automated matching	Performance affected in crowded surveillance environments
2024	Effective Face Recognition from Video Using ESCO-Based Deep CNN Technique [13]	ESCO-optimized deep CNN architecture with frame selection mechanism	Reduced computational overhead and improved video recognition efficiency	Increased training complexity and hyperparameter dependency
2024	Identifying Missing Individual Using Artificial Intelligence [5]	k-NN classification with pretrained face recognition models	Simple implementation and suitable for small datasets	Poor scalability and limited real-time performance
2023	Advancements in Real-Time Face Recognition Algorithms for Enhanced Smart Video Surveillance [15]	Edge-assisted HC-CNN surveillance architecture	High processing speed (>30 FPS) and better scalability	Requires distributed infrastructure and edge-cloud synchronization
2023	Face Detection and Tracking to Find Missing Person Using Haar Cascade Algorithm [6]	Haar Cascade–based face detection and tracking using OpenCV	Low computational cost and easy implementation	Sensitive to lighting, pose variation, and background complexity
2022	AI for Detection of Missing Person (ICAAIC) [7]	Hybrid CNN–SVM framework for feature extraction and classification	Good recognition accuracy and effective feature extraction	Real-time video processing challenges and lighting sensitivity

2021	Missing Child Identification Using Deep Learning [8]	Haar-Cascade detection with LBPH face recognition	Simple architecture and low hardware requirements	Low recognition accuracy under occlusion and poor illumination
2021	Cloud-Based Missing Person Identification System [9]	Cloud-assisted facial recognition with centralized database and alerts	Scalable architecture and rapid notification generation	Dependency on third-party cloud services and privacy concerns
2020	Design and Evaluation of a Real-Time Face Recognition System Using CNNs [14]	CNN-based real-time recognition pipeline	High recognition accuracy with optimized processing	High computational requirements for continuous video streams
2019	ArcFace: Additive Angular Margin Loss for Deep Face Recognition [1]	ArcFace embedding model with angular margin loss	High inter-class separability and robust recognition accuracy	Requires large-scale training datasets and GPU acceleration
2017	Performance Analysis of Real-Time Face Detection System Based on Stream Data Mining Frameworks [12]	Distributed stream processing using Apache Storm and IBM Streams	Supports scalable multi-camera real-time processing	Complex distributed system deployment
2016	Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks (MTCNN) [4]	Cascaded CNN framework for face detection and alignment	Accurate facial localization and landmark detection	High computational latency in large-scale surveillance
2016	Face2Face: Real-Time Face Capture and Reenactment of RGB Videos [10]	Dense 3D facial tracking and GPU-accelerated optimization	Real-time facial modeling and tracking (~28 FPS)	Primarily focused on reenactment rather than surveillance recognition

CHAPTER 3

Problem Statement

To develop an efficient and automated solution for identifying and locating missing persons, addressing the limitations of traditional methods such as poster distribution, public announcements, and manual CCTV review. The absence of a centralized database and reliance on manual comparison hinder timely investigations. An AI-driven surveillance system with real-time facial recognition is required to enhance accuracy, reduce response time, and support law enforcement in reuniting missing individuals with their families.

CHAPTER 4

Software Requirement Specification (SRS)

4.1 Functional Requirements

Functional requirements describe the features and operations implemented in the proposed system for missing person registration, surveillance monitoring, face recognition, and alert generation.

4.1.1 Police Authentication Module

- The system shall allow a police officer to register a new police station using pre-approved station email.
- The system shall authenticate police officers using username and password.
- The system shall allow only authorized officers of a registered station to access the police dashboard.
- The system shall allow police officers to update their login credentials.

4.1.2 Missing Person Registration Module

- The system shall collect officer credentials (name, region number, employee ID, phone number, password).
- The system shall allow officers to select State, District, Taluka, and Police Station.
- The system shall auto-fill the pre-approved police station email from the database.
- The system shall verify the email and create a single account per police station.

4.1.3 Image Preprocessing Module

- The system shall allow police officers to register a missing person complaint.
- The system shall collect guardian information (name, relationship, phone, email, Aadhaar, address).
- The system shall collect missing person details (name, age, DOB, gender, height, weight, hair/eye color, clothing, special marks).
- The system shall allow entry of last-seen location, date, and time.
- The system shall allow uploading up to 5 photos (front, side, full body).
- The system shall store all complaint details in a centralized SQLite database.

4.1.4 Face Detection Module

- The system shall send a confirmation email to the guardian after complaint registration.
- The system shall send alert emails to the concerned police station when AI detects the missing person.
- The system shall log all notifications for audit purposes.

4.1.5 Facial Feature Extraction Module

- The system shall allow police officers to view all registered cases for their station.
- The system shall allow filtering cases by status (Pending, Verified, Closed).
- The system shall allow sorting cases by date (Newest First, Oldest First) and urgency.
- The system shall allow updating case status (Pending → Verified → Closed).
- The system shall allow deletion of a case if required.
- The system shall allow exporting case details as a printable PDF.

4.1.6 Face Matching Module

- The system shall display active missing persons on the homepage.
- The system shall allow the public to view last-seen information, age, and gender of active cases.
- The system shall provide a "Check Status" feature for guardians to track case progress.
- The system shall allow guardians to check status using their Complaint ID.

4.1.7 Live Surveillance Monitoring Module

- The system shall continuously capture frames from connected CCTV cameras.
- The system shall detect faces in frames using RetinaFace.
- The system shall generate face embeddings using ArcFace.
- The system shall compare embeddings with stored missing-person embeddings using cosine similarity.
- The system shall identify a match when similarity exceeds the threshold.
- The system shall generate a detection alert containing:

- Camera ID
- Timestamp
- Missing Person ID
- Snapshot of detected frame
- The system shall send this alert to the corresponding police station automatically.

4.1.8 Alert Generation Module

- The system shall store police station details and authentication data.
- The system shall store all complaints, images, embeddings, and status updates in SQLite.
- The system shall securely store extracted embeddings from ArcFace for faster matching.
- The system shall maintain a log of all alerts detected by the AI module.

4.1.9 Dashboard and Case Management Module

- The system shall ensure that only authenticated police officers can manage cases.
- The system shall restrict modification of records based on station authorization
- The system shall validate all user inputs to prevent invalid or malicious data.
- The system shall securely handle and store uploaded photos.

4.1.10 Database Management Module

- The system shall process CCTV frames in real time or near-real time.
- The AI module shall generate alerts within acceptable latency after detection.

4.2 Non-Functional Requirements

The non-functional requirements define the quality, constraints, and overall operational characteristics of the system. These requirements ensure that the system performs efficiently, remains secure, scalable, and user-friendly throughout its lifecycle.

4.2.1 Performance Requirements

- The system should perform real-time or near real-time face recognition.
- The system should generate alerts with minimal delay after successful detection.
- The recognition pipeline should efficiently process continuous surveillance frames

4.2.2 Reliability Requirements

- The system should provide stable operation during continuous surveillance monitoring.
- The system should maintain accurate storage and retrieval of facial embeddings and case records.
- The system should reduce false detections during recognition operations.

4.2.3 Security Requirements

- The system should restrict unauthorized access to police records and surveillance data.
- The system should authenticate users before dashboard access.
- The system should securely store missing person information and facial embeddings.

4.2.4 Scalability Requirements

- The system should support future expansion with additional surveillance cameras and case records.
- The database structure should support increasing amounts of facial embeddings and alert logs.

4.2.5 Usability Requirements

- The system interface should be simple and user-friendly for police personnel.
- The dashboard should provide easy navigation for complaint registration and monitoring.
- The system should provide clear display of alerts and recognition results.

4.2.6 Maintainability Requirements

- The system should support future updates and improvements in recognition models.

- Individual modules should be maintainable without affecting the overall system operation.
- The codebase should remain modular and organized for easier debugging and enhancement.

4.3 Software Requirements

Table 4.3.1: Software Requirements

Component	Specification
Operating System	Windows 10/11
Programming Language	Python
Framework	Django
Database	SQLite
Frontend Technologies	HTML, CSS, JavaScript, Bootstrap
Backend Technologies	Django Framework
AI Models	RetinaFace, ArcFace
Computer Vision Library	OpenCV
Development Environment	VS Code
Version Control	GitHub

4.4 Hardware Requirements

Table 4.3.1: Hardware Requirements

Component	Specification
Processor	Intel i5 or above
RAM	Minimum 8 GB
Storage	256 GB SSD or above
Camera	Webcam / CCTV Camera
Internet Connection	Required for notifications and remote access

CHAPTER 5

System Design

5.1 System Flowchart

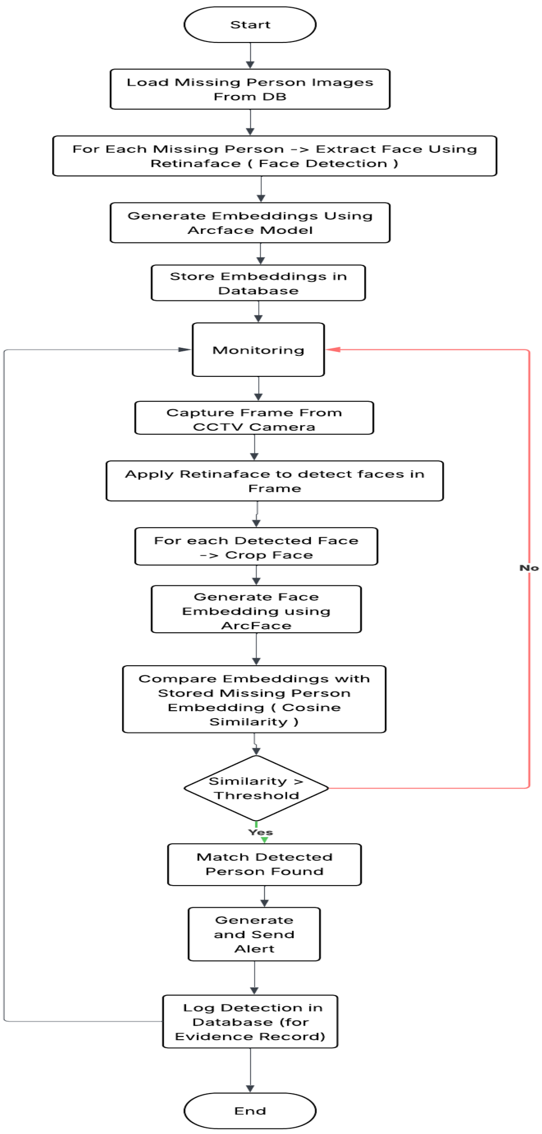

Fig. 5.1 Flowchart

The system flowchart represents the complete workflow of the proposed missing person tracking system. The system captures live CCTV frames, detects faces using RetinaFace, and generates facial embeddings using ArcFace. The generated embeddings are compared with stored database embeddings using cosine similarity. If a valid match is found, the system generates an alert and stores the detection details for further investigation.

5.2 Proposed System Architecture

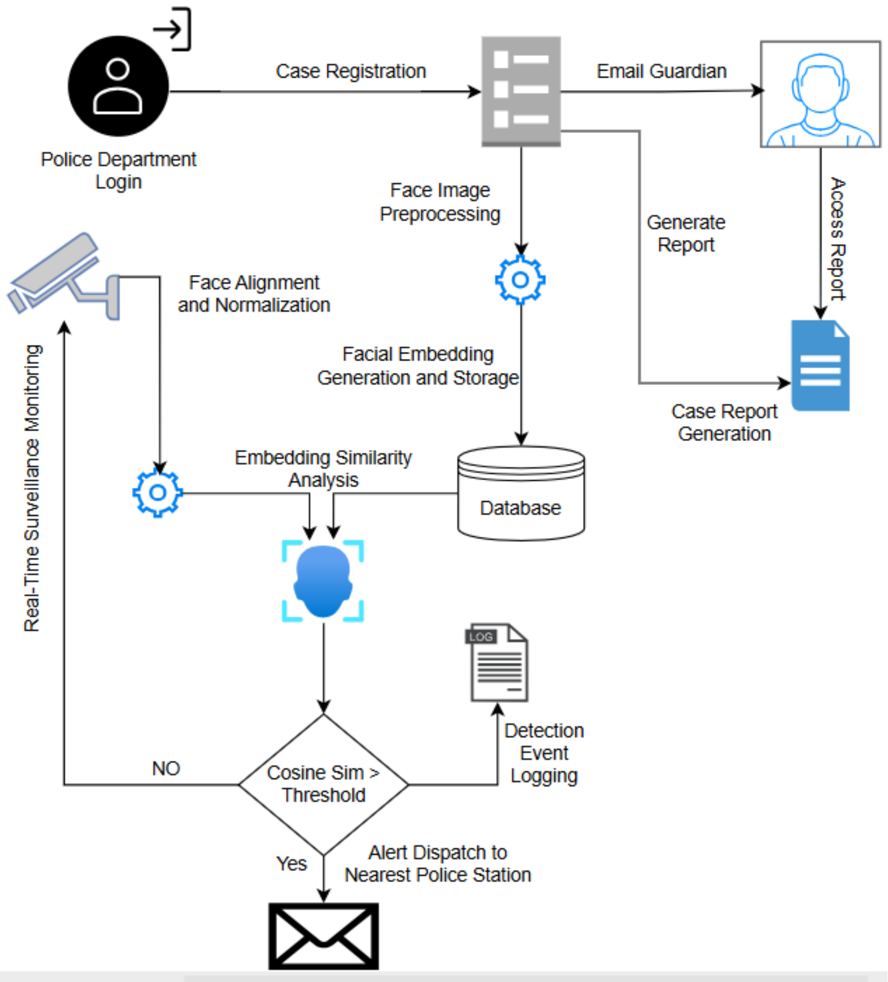

Fig 5.2 System Architecture

The proposed system architecture is designed to automate missing person identification using facial recognition and real-time surveillance monitoring. Initially, police authorities register missing person details and upload facial images into the system. The uploaded images are preprocessed, and facial embeddings are generated using deep learning models and stored in the database.

The system continuously monitors live CCTV footage and detects faces from surveillance streams. The detected faces are compared with stored embeddings using cosine similarity analysis. If the similarity score exceeds the predefined threshold, the system generates an alert notification and sends the detected information to the nearest police station. The architecture also supports report generation and case monitoring for efficient investigation management.

Data Flow Diagrams (DFD)

5.3.1 DFD Level 0

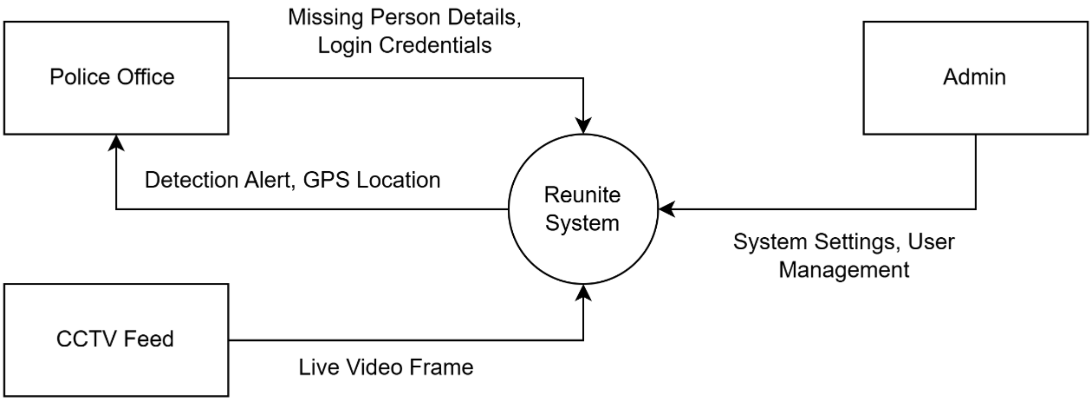

Fig.5.3.1 DFD Level 0

The Level 0 Data Flow Diagram represents the overall functionality of the Reunite system and the interaction between external entities and the main system. The Police Department registers missing person cases by providing personal details and facial images, while the Admin manages system settings and police station information. The system continuously receives live video frames from CCTV feeds for surveillance monitoring. After processing and facial recognition, the system generates detection alerts along with GPS location details and sends them to the concerned police department for further investigation.

5.3.2 DFD Level 1

The Level 1 Data Flow Diagram provides a detailed view of the major functional modules within the Reunite system. It includes modules such as User Profile Management, AI Detection and Recognition, Alert and Notification Engine, Report and Log Management, and Police Station Management. Missing person details and facial embeddings are stored in the database during case registration. Live CCTV video frames are processed through the AI recognition module, and matched detections are forwarded to the alert engine. Detection logs and reports are maintained for monitoring and future investigation purposes.

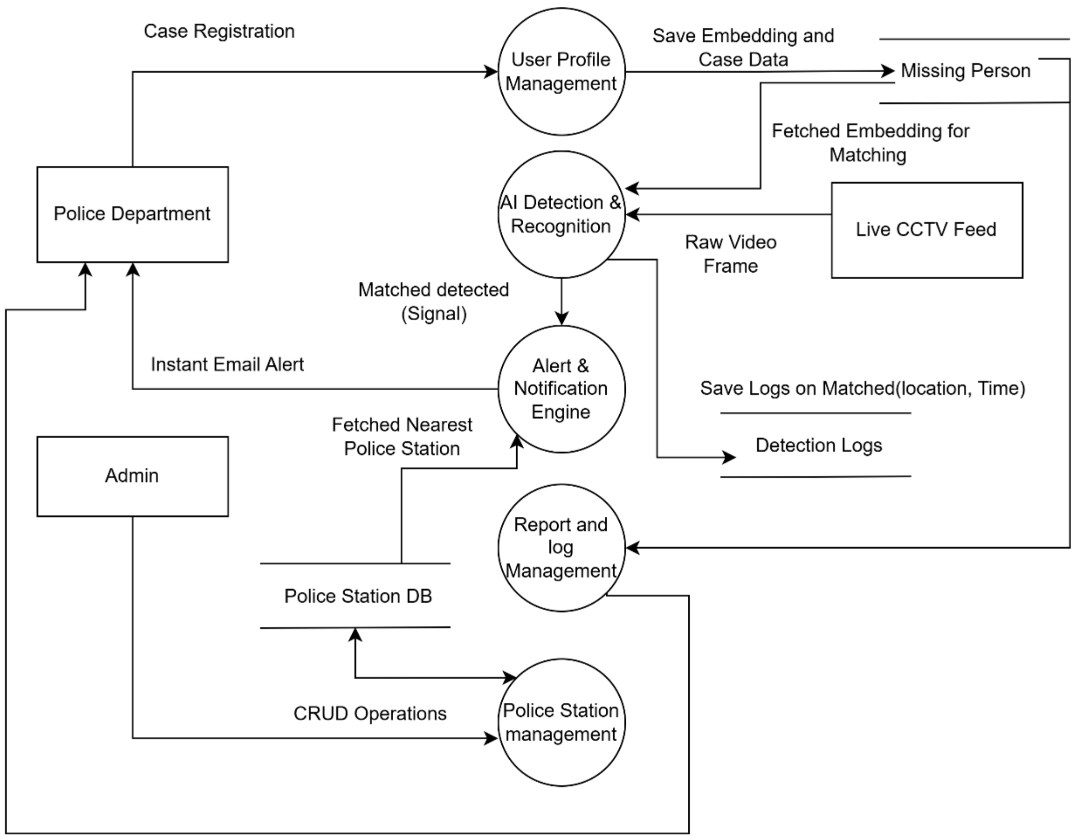

Fig. 5.3.2 DFD Level 1

5.3.3 DFD Level 2

The Level 2 Data Flow Diagram explains the detailed working of the AI-based face recognition process. Live CCTV video frames are first processed using RetinaFace for accurate face detection. The detected faces then undergo image preprocessing and alignment to improve recognition quality. ArcFace is used to generate facial embeddings, which are compared with stored embeddings from the missing person database using comparison logic and cosine similarity analysis. If the similarity score exceeds the predefined threshold, the system identifies a potential match, generates an instant alert notification, and stores detection logs containing location and time information for further investigation.

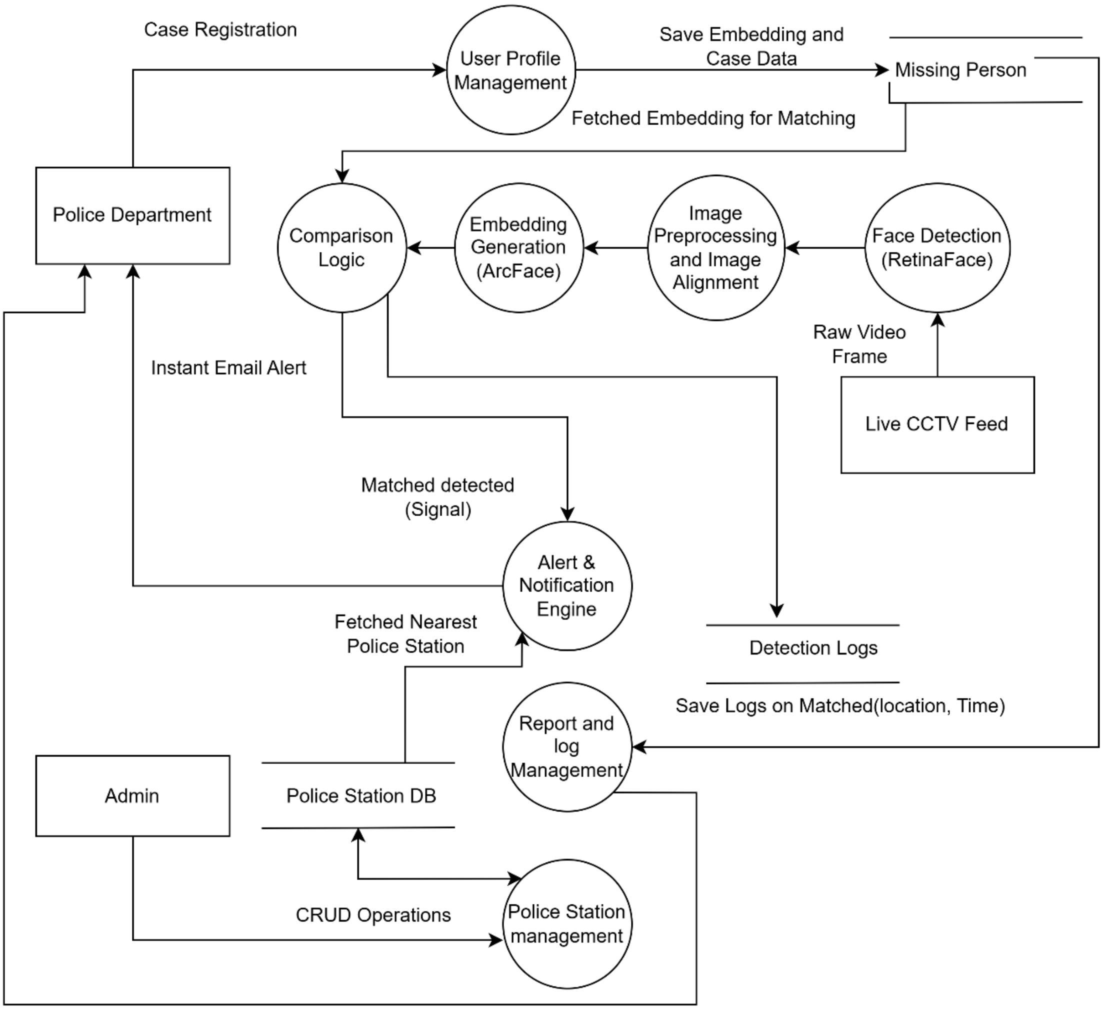

Fig. 5.3.3 DFD Level 2

5.4 UML Diagrams

5.4.1 Use Case Diagram

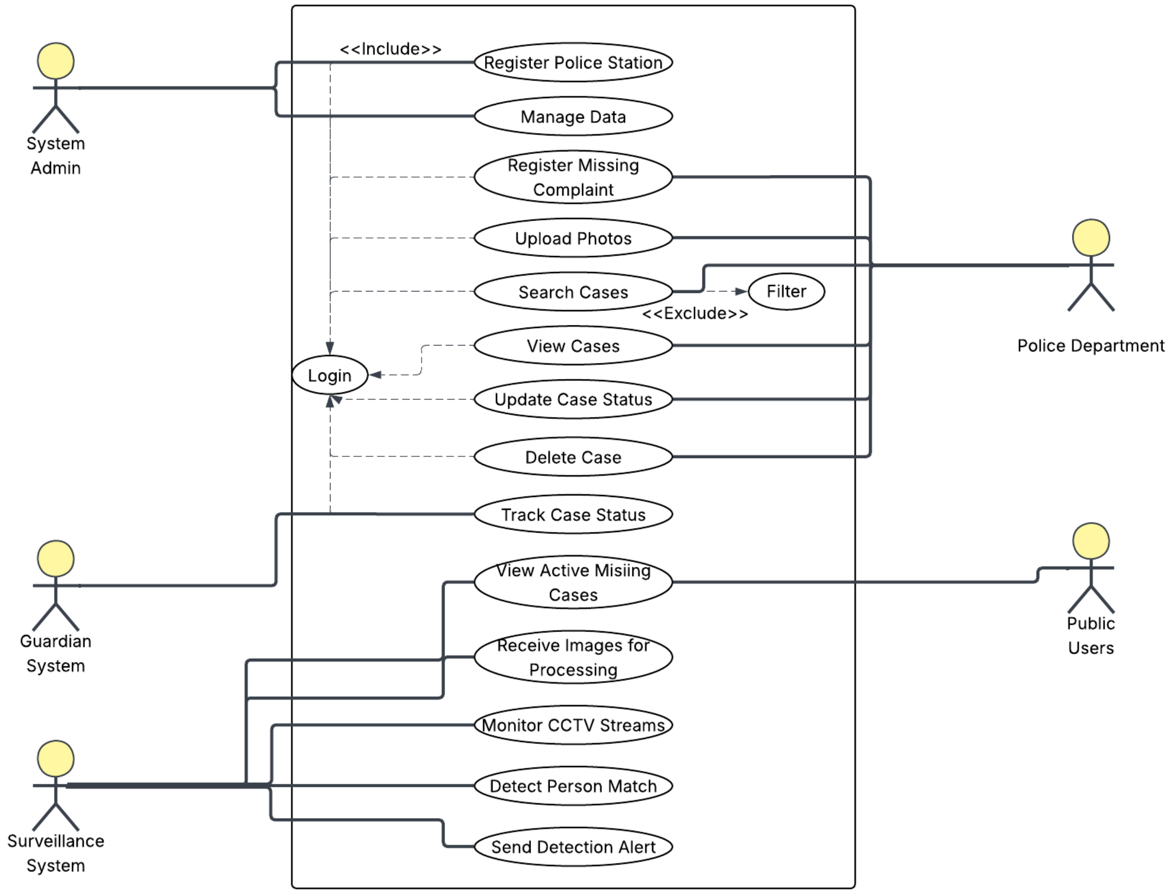

Fig. 5.4.1 Use case diagrams

This Use Case Diagram illustrates the interaction between the System Admin, Police Department, Guardian System, Surveillance System, and Public Users. It showcases core functions such as registering police stations, managing data, filing missing complaints, and monitoring CCTV feeds. The system automates case tracking, facial recognition, and alert generation for efficient missing person detection. Each actor performs defined roles to ensure real-time coordination and swift response.

5.4.2 Class Diagram

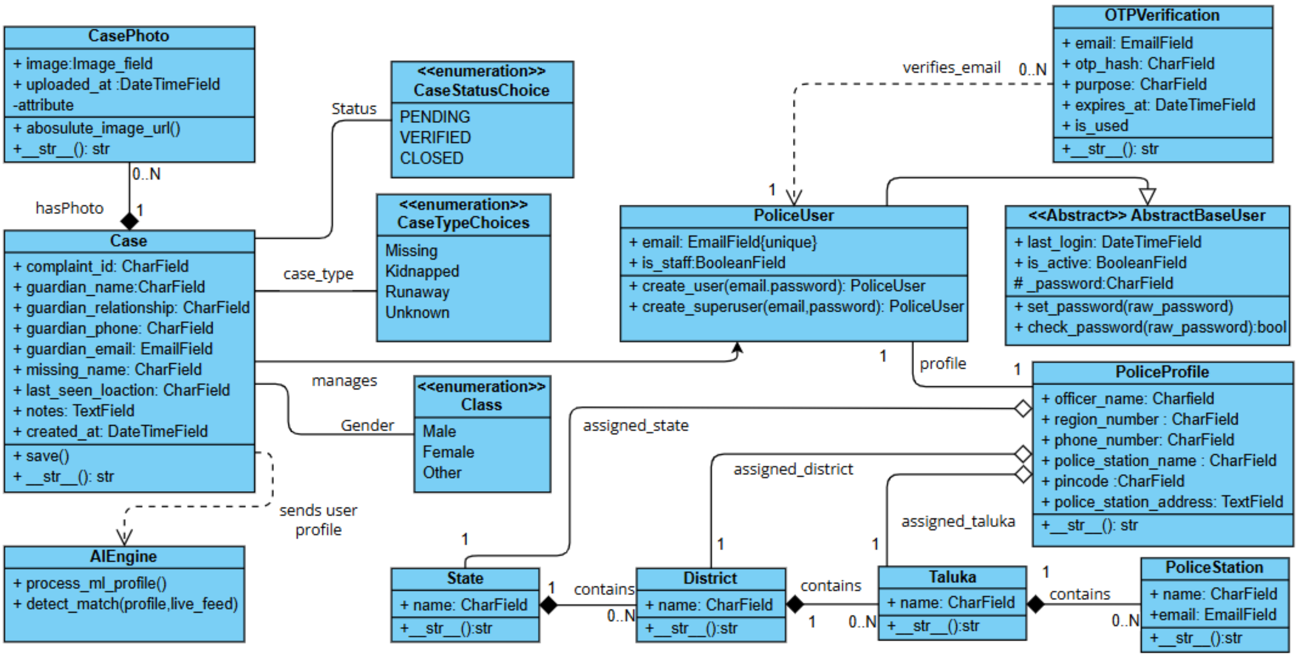

Fig. 5.4.2 Class Diagram

The class diagram shows the working of a missing person detection system using CNN-based facial recognition. It defines relationships between police users, police stations, and missing person cases stored in a centralized database. Each case contains details, guardian info, and multiple photos linked to the AI engine for facial matching. When a match is detected through CCTV or live feed, the system generates alerts and updates the case status.

5.4.3 Sequence Diagram

The sequence diagram illustrates the interaction between different system components during missing person registration and real-time surveillance monitoring. Initially, the police officer logs into the Django web server and uploads missing person details along with facial images. The AI engine processes the uploaded images using RetinaFace for face detection and ArcFace for embedding generation, after which the embeddings are stored in the database.

During live surveillance, video frames from CCTV feeds are continuously processed by the AI engine for face detection and similarity matching with stored embeddings. If the similarity score exceeds the predefined threshold, the system triggers an alert event and dispatches asynchronous notification tasks through Redis Broker and Celery Worker. Finally, an emergency email containing alert details and detected images is sent to the nearest police station through the SMTP email server. If no match is found, the system continues real-time surveillance scanning.

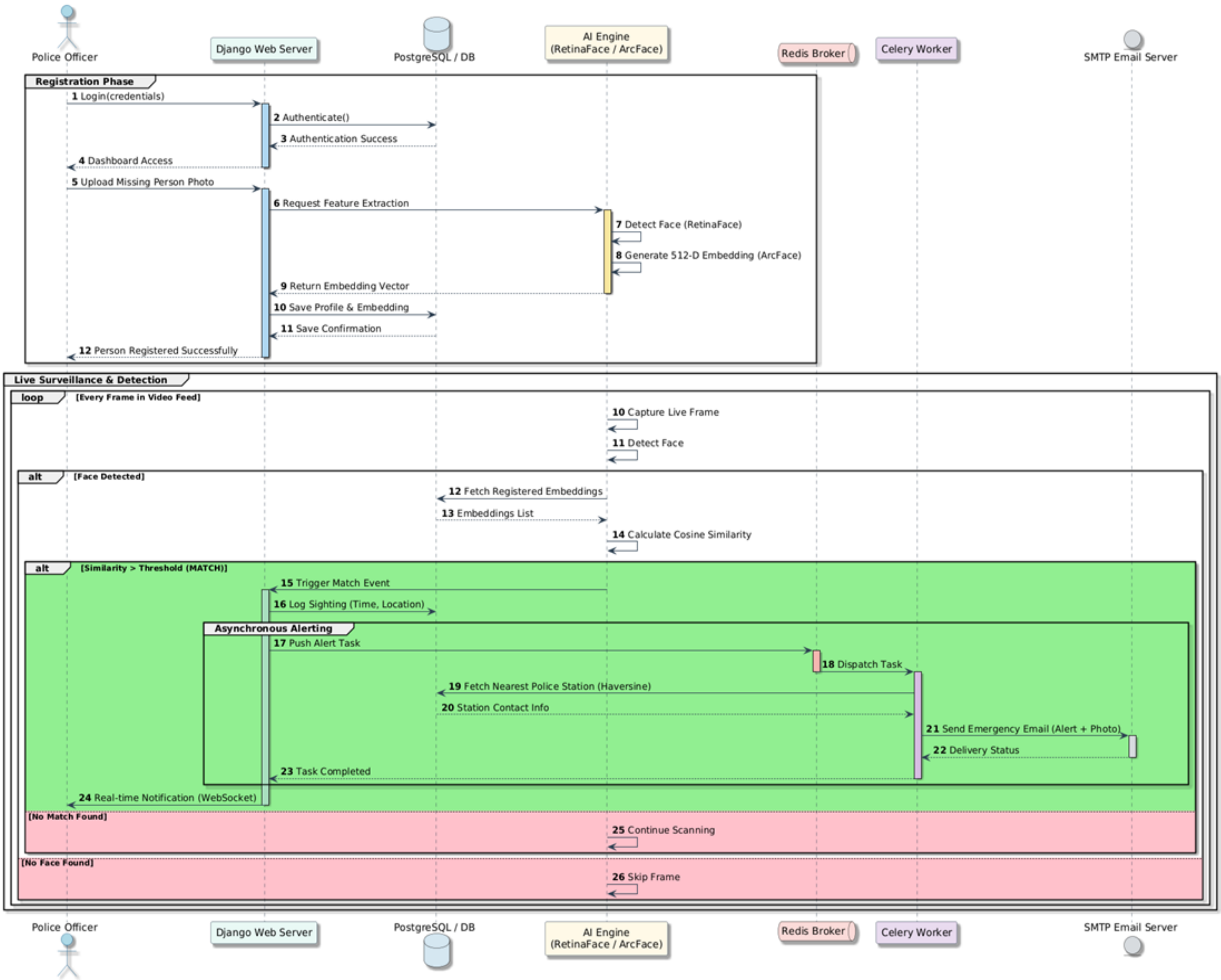

Fig 5.4.3 Sequence Diagram

5.5 Entity-Relationship (ER) Diagram

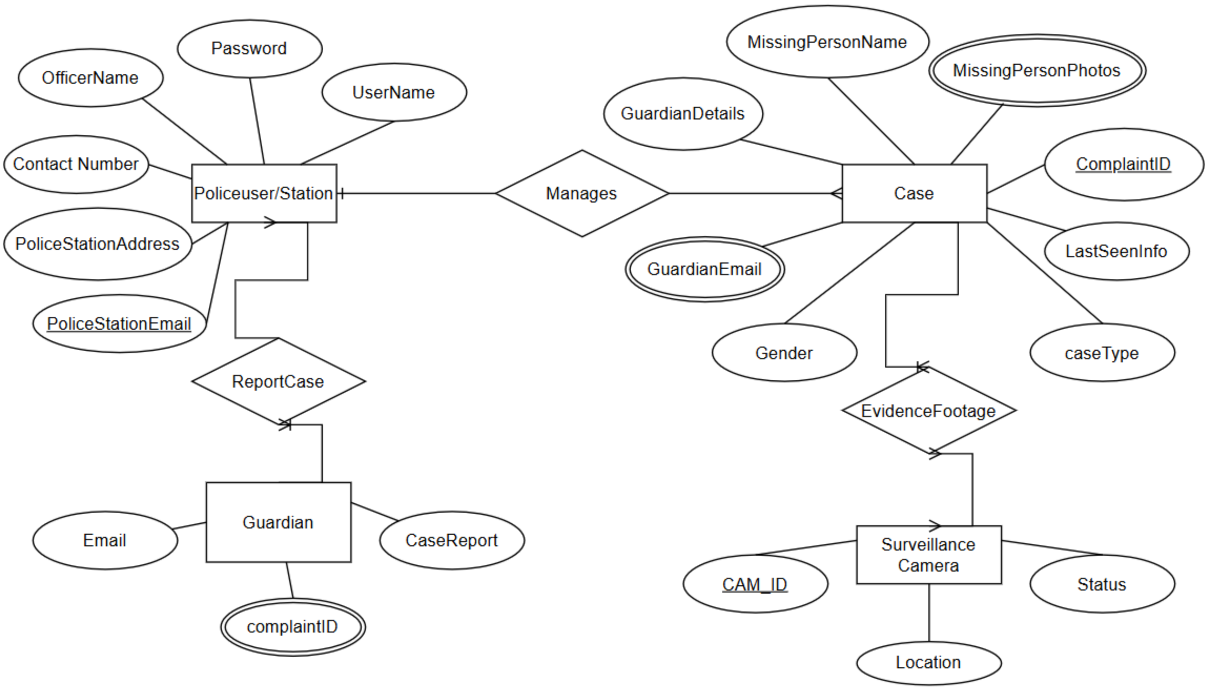

Fig 5.5.1 ER Diagram

The ER diagram represents a missing person tracking and reporting system. Police users manage missing person cases containing details, guardian info, and photos. Guardians can report cases, which are linked to police stations for investigation and updates.

Surveillance cameras provide evidence footage and status updates to help locate the missing person.

CHAPTER 6

System Implementation

6.1 Algorithm

Step 1: Start the system.

Step 2: Guardian reports a missing person case at the nearby police station.

Step 3: Police officer collects missing person details such as personal information, last-seen details, and recent photographs.

Step 4: Police officer registers the case through the Django web application.

Step 5: The system validates the entered information and stores it in the SQLite database.

Step 6: A unique Case ID is generated for the registered case.

Step 7: The system sends an acknowledgment email to the guardian containing registration details and Case ID.

Step 8: The system continuously captures live CCTV or surveillance video streams.

Step 9: RetinaFace detects human faces from each video frame.

Step 10: The detected face undergoes preprocessing, alignment, and normalization.

Step 11: ArcFace generates facial embeddings for the detected face.

Step 12: The generated embedding is compared with stored embeddings using cosine similarity.

Step 13: If the similarity score exceeds the predefined threshold, the system identifies a potential match.

Step 14: An alert notification containing location, timestamp, detected image, and Case ID is sent to the nearest police station.

Step 15: Police officers verify the match and continue the investigation process.

Step 16: After successful identification, the case status is updated to “Closed”.

Step 17: The system sends a confirmation email to the guardian and archives the case record.

Step 18: End.

6.3 Detailed Methodology

This section describes the step-by-step methodology adopted in Reunite: Smart Surveillance for Missing Person Tracking. The workflow begins with case registration by police personnel and proceeds through image preprocessing, face detection, feature extraction, matching, alert generation, verification, and case closure. The system uses RetinaFace for detection and ArcFace for recognition.

6.3.1 Case Registration

The missing person case is registered by authorized police personnel through the Django-based web application. Personal details, last-seen information, and multiple facial images are uploaded and stored securely in the database. The system generates a unique Case ID and sends an acknowledgment email to the guardian.

6.3.2 Image Preprocessing

Uploaded facial images are resized, normalized, and aligned before recognition. Image preprocessing improves the quality and consistency of facial data used during embedding generation and matching.

6.3.3 Face Detection using RetinaFace

The system continuously processes CCTV video streams and detects faces using the RetinaFace model. RetinaFace provides accurate face localization and landmark detection even under varying lighting conditions and facial orientations.

6.3.4 Facial Feature Extraction using ArcFace

After face detection, the aligned facial image is passed to the ArcFace model to generate high-dimensional facial embeddings. These embeddings uniquely represent facial features and are stored for matching operations.

6.3.5 Face Matching using Cosine Similarity

The generated embeddings from live surveillance frames are compared with stored embeddings from the missing person database using cosine similarity analysis. If the similarity score exceeds the predefined threshold value, the system considers it a potential match

6.3.6 Alert Generation and Notification

When a valid match is detected, the system automatically generates an alert notification containing the detected image, Case ID, location details, timestamp, and confidence score. The notification is sent to the nearest police station through email and dashboard alerts.

6.3.7 Verification and Case Management

Police officers verify the detected match and continue further investigation. If the missing person is successfully identified, the case status is updated to “Closed” and the system maintains logs for future reference.

6.4 Protocols Used

In the proposed system, various network and communication protocols are employed to ensure secure, efficient, and reliable data transmission between system components. These protocols facilitate interactions among the web application, database, email server, and surveillance modules

1. **HTTPS (Hypertext Transfer Protocol Secure)**

- Purpose**:** Used for communication between the client (web browser) and the Django web server.
- Description**:**
    - All user interactions such as case registration, status updates, and data retrieval occur through HTTP requests and responses.
    - The secure variant, HTTPS, is implemented to encrypt all transmitted data, ensuring privacy and protection against unauthorized access or interception.
- Role in System: Enables secure access to web pages, API endpoints, and data exchange between client and server.

2. **SMTP (Simple Mail Transfer Protocol)**

- Purpose**:** Handles the delivery of email notifications between the web application and guardians or police officials.
- Description:
    - After case registration, the system uses SMTP to send acknowledgment and confirmation emails.
    - Alerts generated upon face matches are also dispatched via SMTP to the concerned authorities.
- Role in System: Provides automated, reliable, and real-time communication to notify users about registration, detection alerts, and case closure.

CHAPTER 7

GUI and Experimental Results

7.1 Testing Strategy

The proposed system was tested using unit testing, functional testing, and acceptance testing techniques to verify the correctness and reliability of individual modules and overall system functionality.

1. Unit Testing

Unit testing was performed on individual modules such as user authentication, case registration, image upload, database operations, email notification, and face matching functions. Each module was tested independently using valid and invalid inputs to verify proper functionality and error handling.

Examples:

- Login validation with correct and incorrect credentials
- Missing person registration with mandatory field checking
- Image upload validation
- Database insertion and retrieval testing

Table 7.1.1 - Login Module Testing

Test Case	Expected Result	Actual Result	Status
Valid Login	Dashboard opens	Successful	Pass
Invalid Password	Error message	Successful	Pass
Empty Fields	Validation message	Successful	Pass

Table 7.1.2 - Missing Person Registration Testing

Test Case	Expected Result	Actual Result	Status
Valid Registration	Case stored	Successful	Pass
Missing Required Fields	Validation error	Successful	Pass
Multiple Image Upload	Images stored	Successful	Pass

2. Functional Testing

Functional testing was carried out to verify whether the complete system performed according to the expected workflow. The interaction between modules such as surveillance monitoring, face detection, embedding generation, similarity comparison, and alert generation was tested using real-time webcam and CCTV video streams.

Examples:

- Detecting faces from live video feeds
- Matching detected faces with stored database embeddings
- Generating alerts when similarity exceeds threshold
- Updating case status after successful identification

Table 7.1.3 - Face Detection and Recognition Testing

Test Case	Expected Result	Actual Result	Status
Face Detected in CCTV	Face identified	Successful	Pass
Unknown Face	No match	Successful	Pass
Low Lighting	Reduced accuracy	Observed	Pass

Table 7.1.4 - Alert Generation Testing

Test Case	Expected Result	Actual Result	Status
Match Found	Alert generated	Successful	Pass
Email Notification	Email delivered	Successful	Pass
No Match	No alert	Successful	Pass

3. Acceptance Testing

Acceptance testing was conducted to ensure that the system satisfied the project objectives and user requirements. The system was tested from the perspective of police personnel using real-world usage scenarios such as case registration, monitoring, and alert verification.

Examples:

- Registering a missing person case
- Receiving alert notifications during successful matching

- Verifying generated reports and case details
- Monitoring real-time surveillance workflow

Optional Table 7.1.5 - Performance Evaluation

Parameter	**Observation**
Face Detection	Accurate under normal lighting
Recognition Speed	Near real-time
Alert Generation	Successful
Database Retrieval	Fast
Surveillance Monitoring	Continuous

7.2 Test Case Selection

Test cases were selected based on critical functionalities of the system. Both normal and exceptional scenarios were considered to evaluate system reliability and performance. The selected test cases included:

- Valid and invalid login attempts
- Registration with incomplete information
- Face detection under different lighting conditions
- Matching with correct and incorrect facial data
- Alert generation for successful matches
- Continuous surveillance monitoring

7.3 System Output

1) Home Page

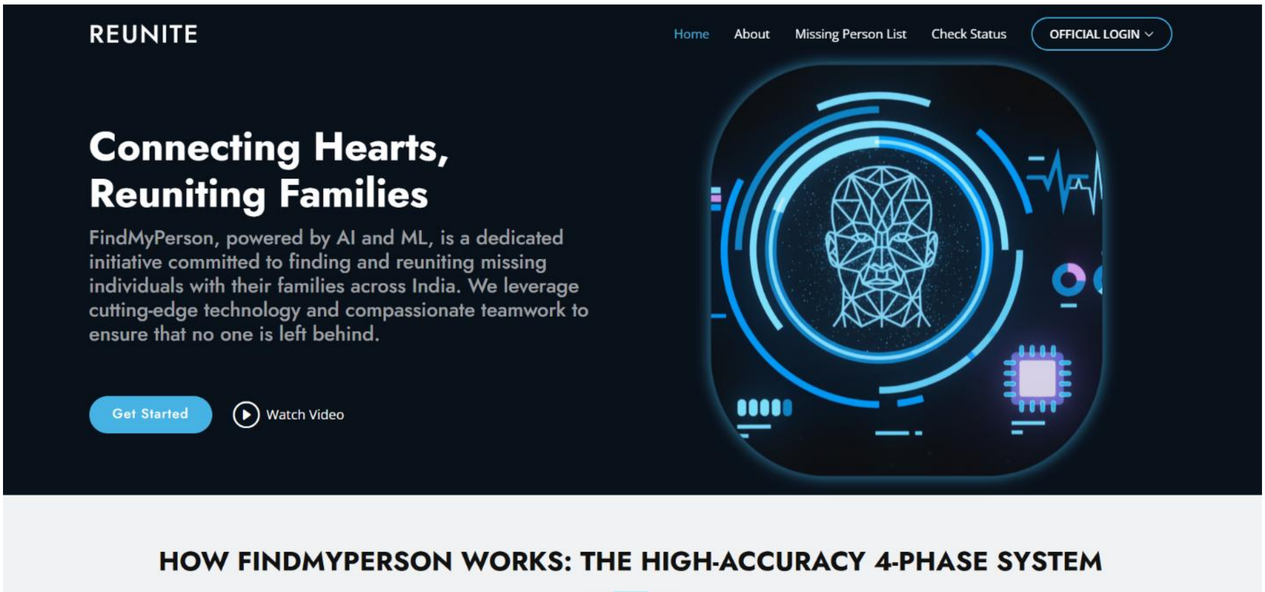

Fig. 7.3.1 Home Page

2) Department Login

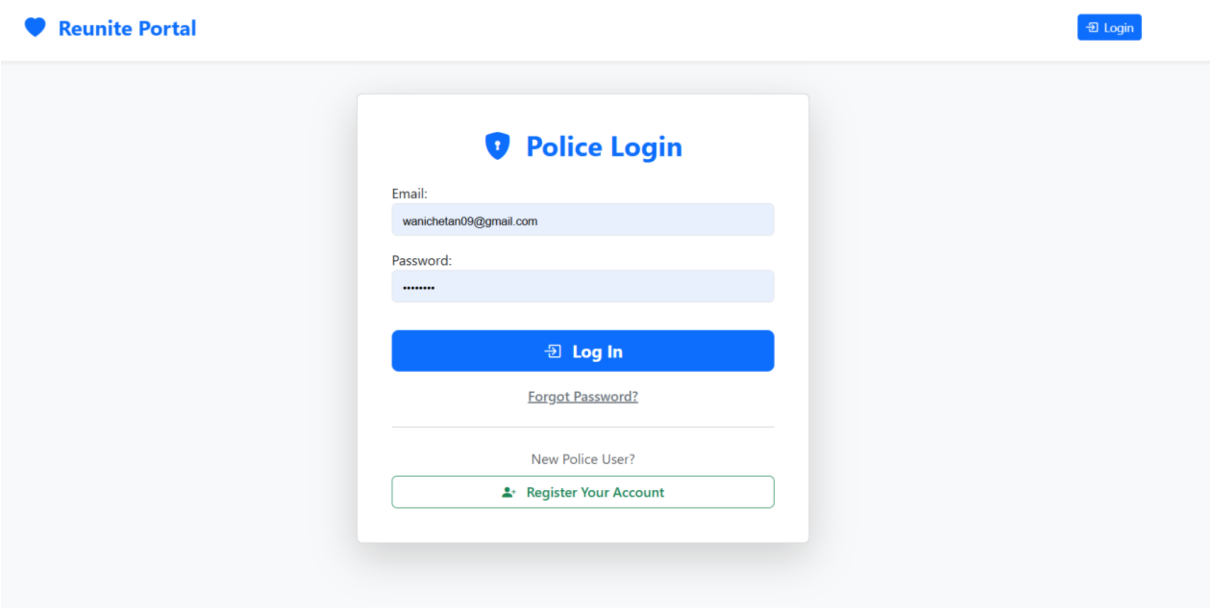

Fig 7.3.2 Department Login

3) Police Dashboard

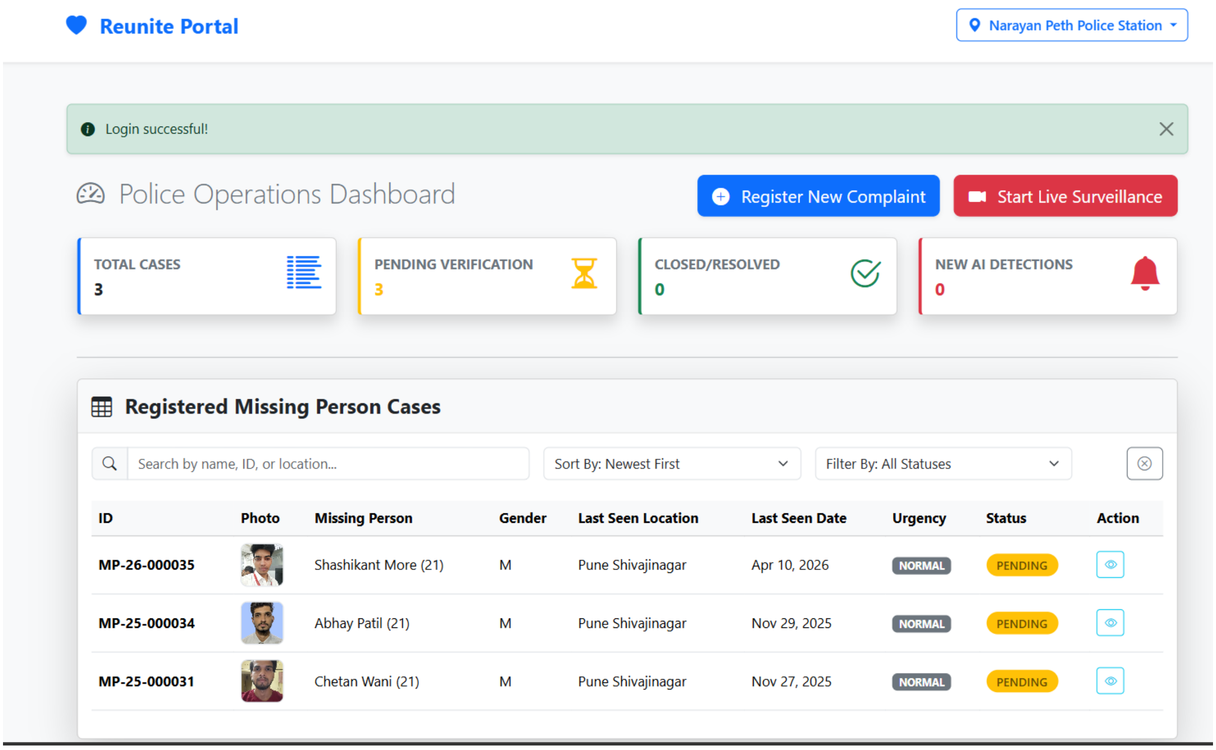

Fig.7.3.3 Police Dashboard

4) Missing Person Registration

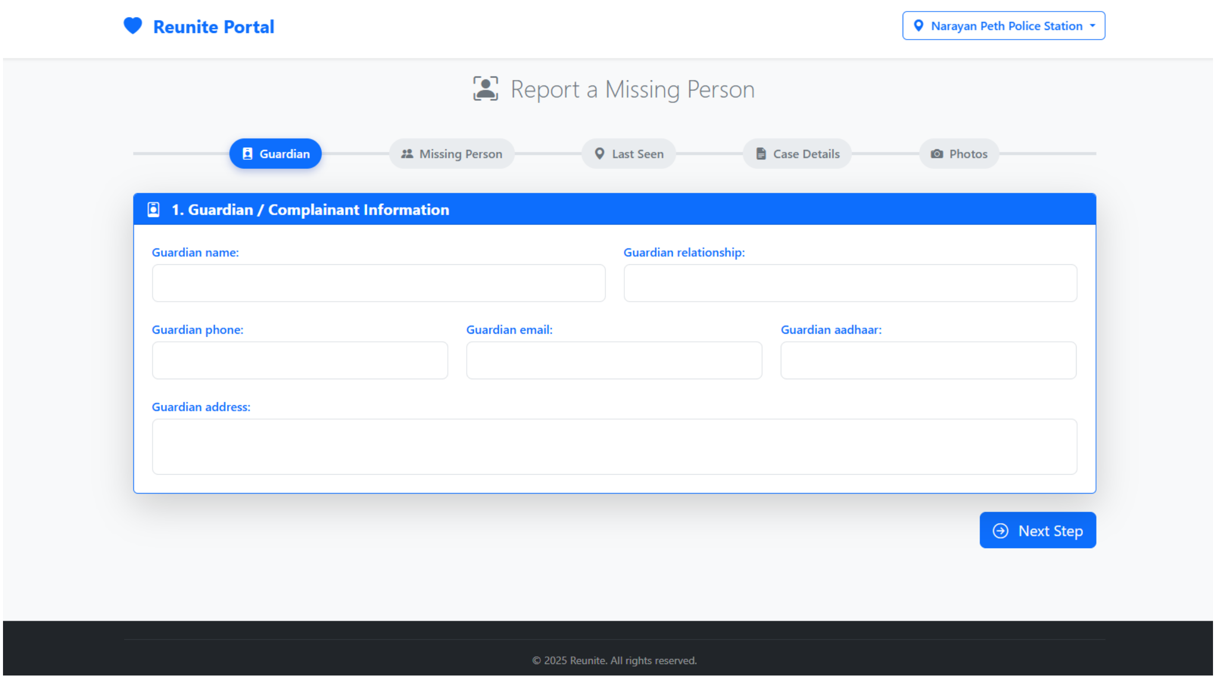

Fig. 7.3.4 Missing Person Registration

5) Face Detection & Recognition

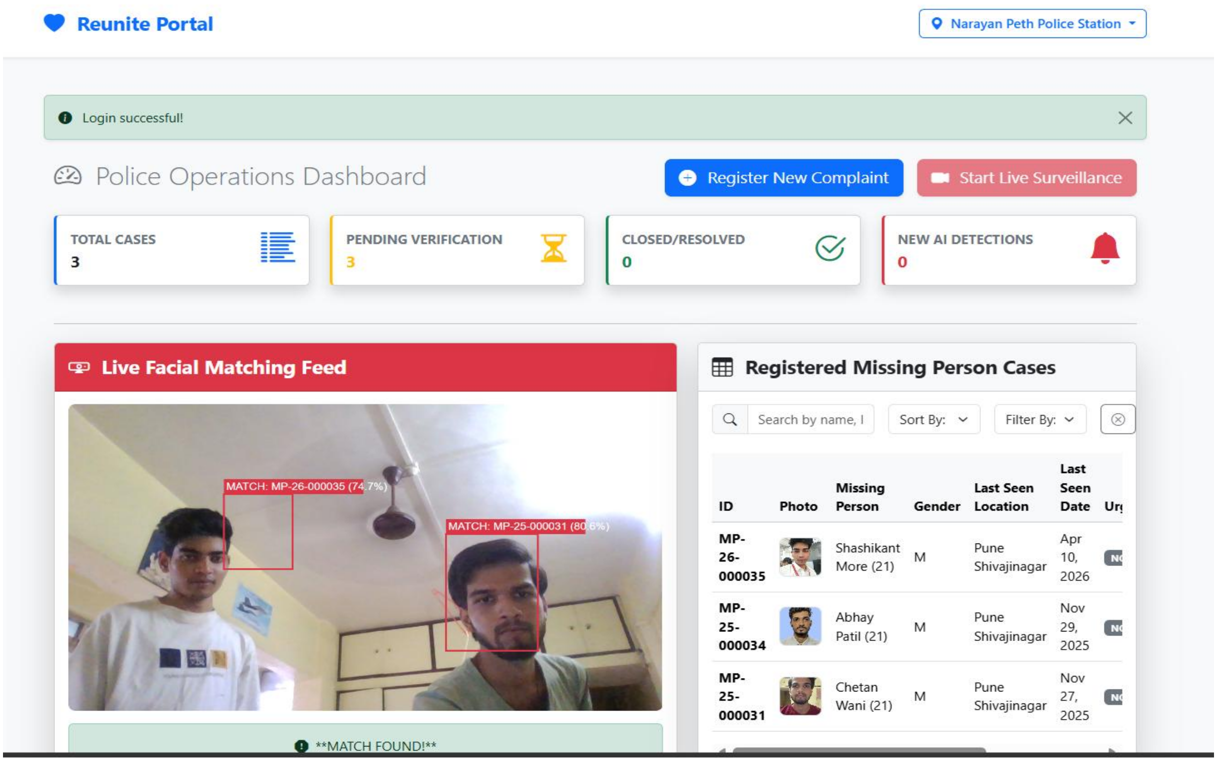

Fig.7.3.5 Face Detection & Recognition

6) Alert Notification Output

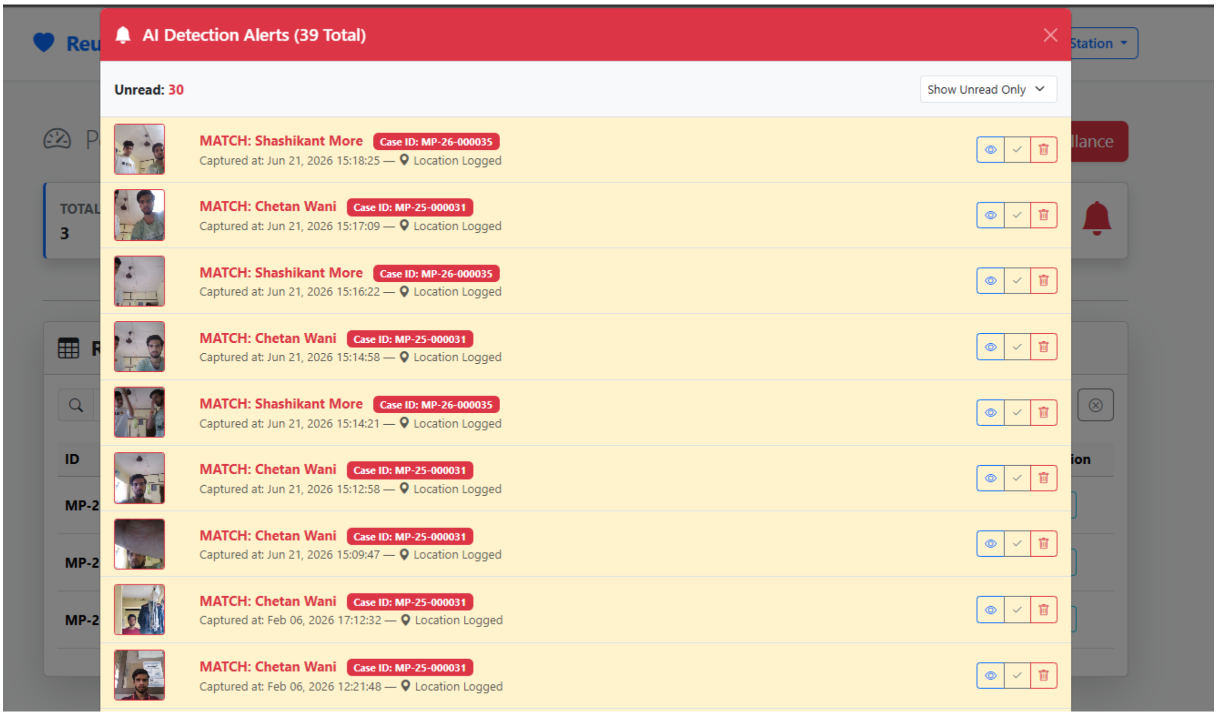

Fig. 7.3.6 Alert Notification Output

7) Case Report Generation

Reunite

CASE ID: MP-25-000019

Reunite Missing Persons Bureau

Government of India • Official Case Document

Missing Person Official Case Report

1. Missing Person Details

Name:	SHASHIKANT MORE		
Gender:	M	**Age / DOB:**	21 / 2004-09-05
Height / Weight:	160 / 50	**Hair / Eye Color:**	Black / Black
Last Known Clothing:	Last seen wearing a dark blue formal shirt and beige trousers.		
Special Identifying Marks:	No identifiable marks		

2. Incident & Location Details

Case Status:	PENDING	**Urgency Level:**	**HIGH**
Last Seen Location:	Narayan Peth Pune, Shivajinagar		
Last Seen Date/Time:	Nov 02, 2025 at 8:21 p.m.	**Case Registration Date:**	Nov 09, 2025 15:06
Case Notes:	Guardian mentioned the person may have gone to visit relatives in a nearby district.		

3. Complainant / Guardian Details

Name:	Kailash More (Father)	**Aadhaar:**	878745847855
Phone:	8858585854	**Email:**	moreshashikant332@gmail.com
Address:	Gat no 41, Plot no 36, Nanded, Maharashtra		

OFFICIAL STAMP Reunite OFFICIAL STAMP

Official Stamp

Reporting Officer's Signature
[Officer Name]

Scan to Verify Document

Disclaimer: This document is generated by the Reunite Missing Persons Bureau's official system and is intended solely for official purposes. While every effort is made to ensure the accuracy of the information, the Bureau does not assume responsibility for any errors, omissions, or misinterpretations. The information contained herein is confidential and may not be reproduced, distributed, or used for any unauthorized purposes. Verification from official sources is recommended before taking any action based on this report. Any unauthorized use, copying, or disclosure of this document is strictly prohibited and may be subject to legal action.

Document Authenticated

Report ID: MP-25-000019 | Report Status: PENDING

Reunite Missing Persons Bureau • This is a system-generated document.

Fig 7.3.7 Case Report Generation

8) Face Embeddings Storage

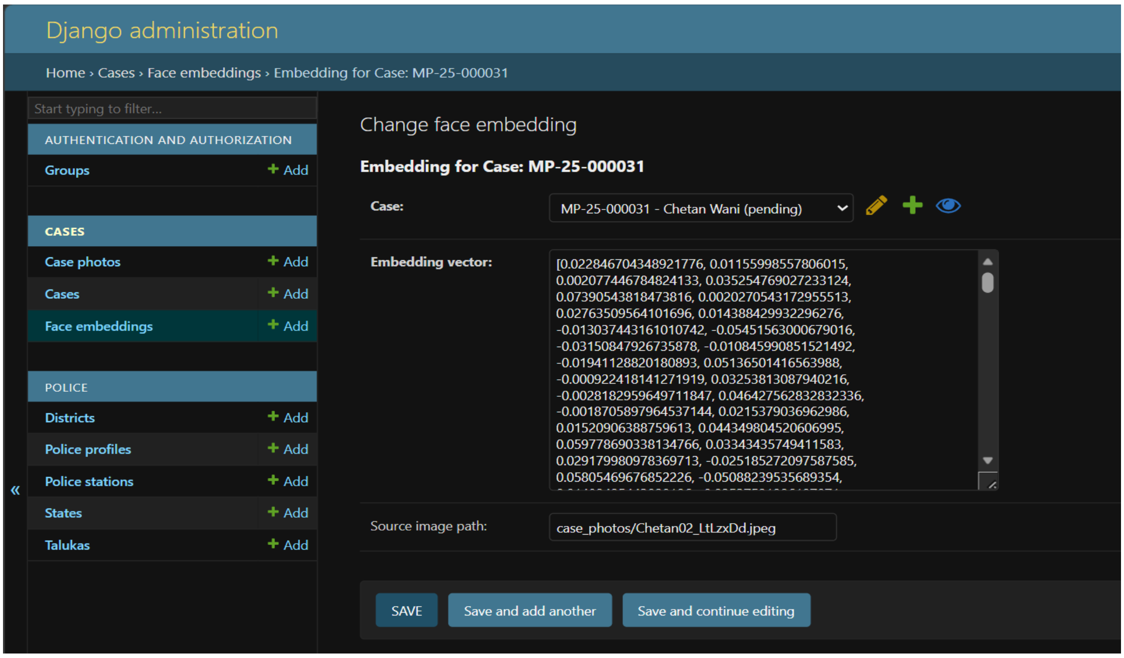

Fig 7.3.8 Face Embeddings Storage

9) Automated Email Notification

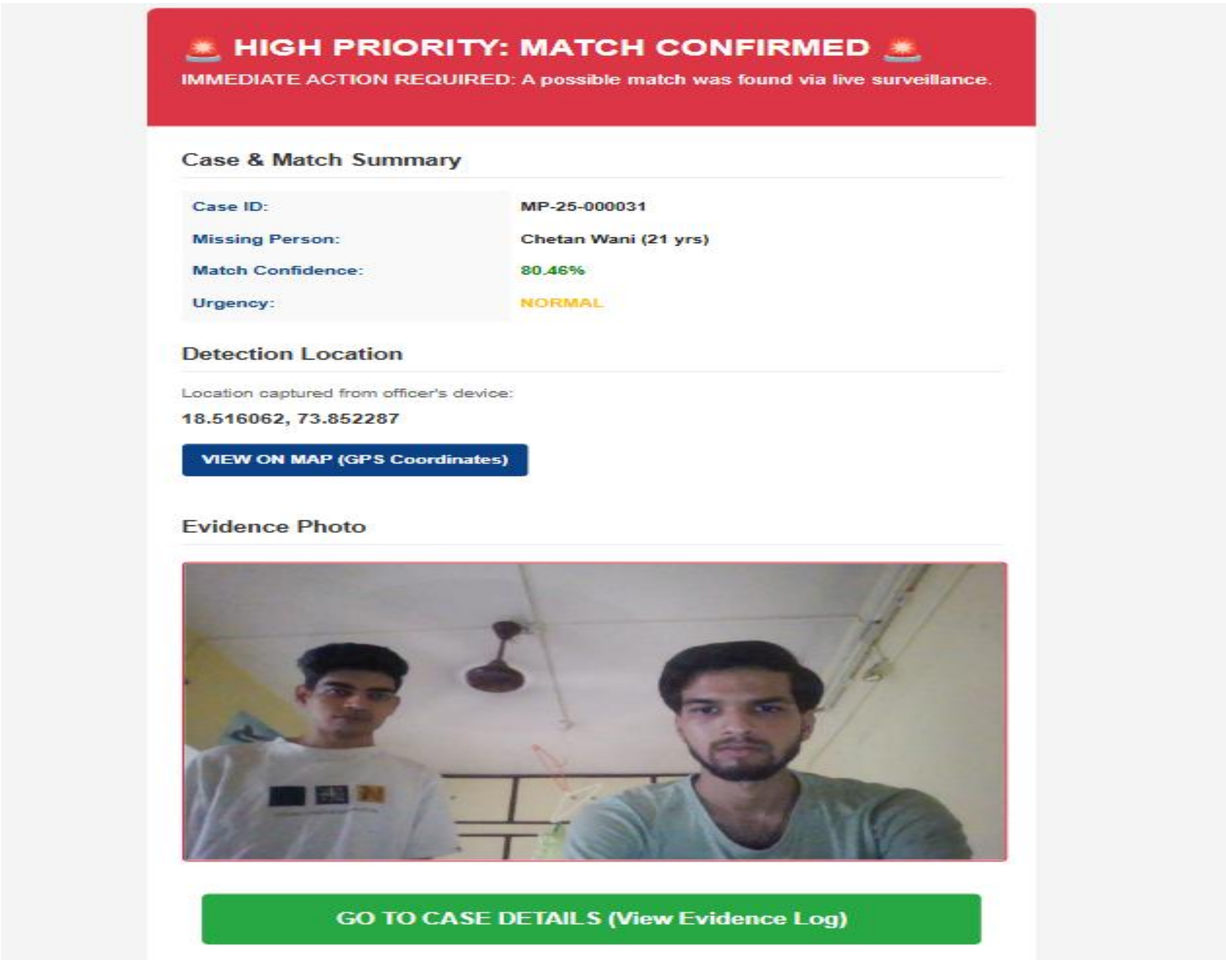

Fig 7.3.9 Automated Email Notification

CHAPTER 8

Conclusions & Future Work

8.1 Observations

During the development and testing of the Reunite system, it was observed that the integration of deep learning and real-time surveillance significantly improved the efficiency of missing person identification. The combination of RetinaFace and ArcFace provided reliable face detection and recognition performance under different surveillance conditions. The system was capable of detecting faces from live CCTV or webcam feeds and comparing them with stored database records in real time.

The Django framework provided an effective platform for handling user authentication, case registration, database management, and alert generation. Automated email notifications and dashboard alerts helped improve communication between police authorities and guardians during investigation procedures.

Experimental testing showed that the system performed efficiently under normal lighting conditions and frontal or moderately angled face positions. The use of cosine similarity for facial embedding comparison enabled faster matching and reduced manual investigation efforts. Overall, the proposed system demonstrated the practical applicability of AI-based surveillance for missing person tracking and case management.

8.2 Limitations

Although the proposed system achieved satisfactory results, certain limitations were observed during implementation and testing:

1. **Lighting and Pose Variations:** Recognition accuracy decreases under poor lighting conditions, blurred video frames, or extreme facial angles.
2. **Occlusion Challenges:** Partial face occlusion caused by masks, helmets, or obstacles may affect face detection and matching performance.
3. **Hardware Dependency:** Real-time face recognition requires adequate processing power, especially during continuous CCTV monitoring.
4. **Scalability Constraints:** The current implementation is suitable for small to medium-scale surveillance environments and may require optimization for large-

scale deployment.

5. **Network and Camera Quality:** System performance depends on camera resolution, frame quality, and stable network connectivity.
6. **Privacy and Security Concerns:** Continuous surveillance and facial data storage require proper security measures and controlled system access.

8.3 Future Work

The proposed system can be further improved by incorporating advanced features and scalable deployment strategies. Future enhancements may include:

- Integration with cloud infrastructure for large-scale surveillance processing and centralized data storage.
- Development of a mobile application for police officers and guardians to receive real-time alerts and case updates.
- Integration with government or national missing person databases for wider identification coverage.
- Improvement in recognition performance under low-light and crowded surveillance conditions.
- Optimization of AI models for faster processing and reduced computational requirements.
- Implementation of stronger encryption and advanced access-control mechanisms for enhanced data privacy and security.
- Support for multi-camera tracking and real-time GPS-based surveillance integration.
- Continuous model training using larger and more diverse facial datasets to improve recognition accuracy.

REFERENCES

[1] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4690–4699, 2019.

[2] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4690–4699, 2019.

[3] R. Sharma, A. Verma, and P. Kulkarni, "Multiscale Facial Detection Using RetinaFace Architecture," *International Journal of Advanced Computer Vision and Applications*, vol. 12, no. 3, pp. 45–52, 2025.

[4] P. S. Rao, K. S. Reddy, and M. V. Kumar, "AI Finder: An Intelligent Framework for Finding Missing Persons Using Face Match Algorithm," *International Journal of Innovative Research in Technology (IJIRT)*, vol. 11, no. 2, pp. 112–118, 2025.

[5] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, 2016.

[6] S. B. Sharath and H. Kumar, "Identifying Missing Individual Using Artificial Intelligence," *International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)*, vol. 4, no. 1, pp. 98–103, 2024, doi: 10.48175/IJARSCT 19215.

[7] G. A. V. R. C. Rao, S. K. Reddy, and A. Kumar, "Face Detection and Tracking to Find Missing Person Using Haar cascade Algorithm," *International Journal for Research Trends and Innovation (IJRTI)*, vol. 8, no. 7, pp. 234–239, 2023.

[8] B. Vinavatani, R. Mehta, and A. Shah, "AI for Detection of Missing Person," in *Proceedings of the International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, IEEE, 2022, pp. 1–6, doi: 10.1109/ICAAIC53929.2022.9792672.

[9] M. Bharathwaj, P. Naveen, and S. Ramesh, "Missing Child Identification Using Deep Learning," *International Journal of Creative Research Thoughts (IJCRT)*, vol. 9, no. 6, pp. 1021–1026, 2021.

[10] S. Pawar, A. Patil, and R. Kulkarni, “Find Missing Person Using Artificial Intelligence,” *International Research Journal of Engineering and Technology (IRJET)*, vol. 8, no. 12, pp. 1564–1568, 2021.

[11] J.Thies, M. Zollhöfer, M. Stamminger, C. Theobalt, and M. Nießner, “Face2Face: real-time face capture and reenactment of RGB videos,” in Proceedings of the IEEE

[12] Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 2387–2395.H. A. Correia, D. Pontes, and J. H. Brito, “Face recognition in online soccer streaming for piracy detection,”Multimedia Tools and Applications, vol. 84, pp. 30737–30756, 2025, doi: 10.1007/s11042 024 20389 3.

[13] N. L. Kazanskiy, V. Protsenko, and P. Serafimovich,“Performance analysis of real-time face detection system based on stream data mining frameworks,” *Procedia Engineering*, vol. 201, pp. 806–816, 2017.

[14] J. C. Musale, A. K. Singh, and S. Shirke, “Effective face recognition from video using enhanced social collie optimization based deep convolutional neural network technique,” Journal of Visual Communication and Image Representation, 2024.

[15] P.K.B.Pranav and J.Manikandan, “Design and evaluation of a real-time face recognition system using convolutional neural networks,” Procedia Computer Science, vol. 171, pp. 1651–1659, 2020.

[16] K. Sa, A. Ra, B. B. S. Krishna, and V. Pa,“Advancements in real-time face recognition algorithms for enhanced smart video surveillance,” Procedia Computer Science, vol. 230, pp. 486–492, 2023.

Appendices

Appendix A: Published Research Paper (First Page) & Publication Certificates

International Research Journal of Innovations in Engineering and Technology (IRJIET)

ISSN (online): 2581-3048

Volume 10, Issue 6, pp 193-200, June-2026

https://doi.org/10.47001/IRJIET/2026.106024

A Survey on Face Detection and Recognition Techniques

[1]Ashwini Bhamre, [2]Chetan K. Wani, [3]Shashikant K. More, [4]Aaradhya S. Sonar

[1]Assistant Professor, Department of Information Technology Engineering, PES Modern COE, Pune, Maharashtra, India
[2,3,4]Student, Department of Information Technology Engineering, PES Modern COE, Pune, Maharashtra, India

***Abstract* - The use of face detection and recognition has emerged as a theme of research in the field of computer vision due to their wide application in security systems, biometrics authentication, access control and intelligent video analysis. The rapid development of visual information produced by surveillance systems and digital imaging systems has raised the need to find automated, scaled up, and robust face recognition methods. This is a contribution that has made a lot of progress over the last ten years following the shift of traditional feature-based techniques involving handcrafted features to deep learning driven detection and embedding based recognition models.**

The paper is a thorough review of face detection methods, face recognition systems, strategies of similarity measurements and real-time surveillance systems. Classical techniques and the contemporary deep learning-based techniques are systematically reviewed and compared. The survey also considers the distributed and large-scale processing structures on one side, and system level issues like real-time performance, scalability, open set recognition and computational efficiency on the other. Judging by the comparative analysis, it is possible to determine the main research gaps and perspectives in the future and offer the systematized vision of the existing developments in face detection and recognition technologies.

Keywords: Face Detection, Face Recognition, Deep Learning, Surveillance Systems, RetinaFace, ArcFace, Facial Embeddings.

I. INTRODUCTION

Face detection and recognition has developed into a high profile area of research in computer vision and pattern recognition because of its broad range of application in security systems, biometric authentication, forensic pattern recognition, access control, and intelligent video surveillance The fast development of imaging equipment and surveillance infrastructure has created massive amounts of visual information that require automated and scalable face recognition algorithms, which can perform at various real world scenarios.

Early face recognition systems were mainly based on handwritten feature extraction schemes with classical classifiers Techniques Haar cascade classifiers [6], Principal Component Analysis (PCA) [7], and statistical appearance based models [8] were common owing to their computational simplicity and capability to run on a limited hardware platform but failed to perform well in an unconstrained settings, including variations in illumination, change in pose, conclusiveness, and low resolution imagery.

Convolutional Neural Networks (CNNs) The advent of deep learning has largely changed face analysis Multistage face detectors like MTCNN [4], single stage face detectors like RetinaFace [2], and embedding based recognition models like ArcFace [1] have both shown strong localization accuracy across scale and open set recognition Multistage detectors Multistage detectors, e.g. MTCNN [4], have demonstrated high localization accuracy across scales; single stage detectors Multistage detectors, e.g. RetinaFace [2], have demonstrated high localization accuracy in Recent work has, however, also been extended to scalable and real-time deployment models including distributed stream processing systems [12], CNN based real-time recognition systems [14], and hybrid edge cloud surveillance models [15] Nonetheless, the variety of existing models including traditional handcrafted models, deep embedding based models, similarity matching models, and system level architecture makes it hard to gain a coherent and systematic view of the discipline.

The purpose of the survey is to offer a critical overview of the face detecting and recognizing techniques The paper reviews critically the current methods of detecting faces with traditional and deep learning approaches, embedding recognition models and similarity strategies of recognition evaluation and the surveillance architecture in real-time By summarizing the current research studies and revealing the significant gaps in understanding the field, the paper will present a systematic overview of the current developments and suggest the important directions of the research.

CERTIFICATE OF PUBLICATION

IRJIET

INTERNATIONAL RESEARCH JOURNAL OF INNOVATIONS IN ENGINEERING AND TECHNOLOGY (IRJIET)

(International Open Access, Monthly, Peer Reviewed Journal)

Is Hereby Awarding this Certificate to

Ashwini Bhamre

Assistant Professor, Department of Information Technology Engineering, PES Modern COE, Pune, Maharashtra, India

In Recognition of the Publication of Manuscript Entitled

A Survey on Face Detection and Recognition Techniques

Published in

International Research Journal of Innovations in Engineering and Technology (IRJIET)

Volume 10, Issue 6, pp 193-200, June-2026

https://doi.org/10.47001/IRJIET/2026.106024

(2026) Impact Factor 8.7

IRJIET

Editor-In-Chief
IRJIET

Managing Editor
IRJIET

ISSN (online): 2581-3048

Mail us at: editor@irjiet.com / irjietjournal@gmail.com
Journal Website : www.irjiet.com

Manuscript ID : IRJIET106024
Date of Issue : June 22, 2026

CERTIFICATE OF PUBLICATION

IRJIET

INTERNATIONAL RESEARCH JOURNAL OF INNOVATIONS IN ENGINEERING AND TECHNOLOGY (IRJIET)

(International Open Access, Monthly, Peer Reviewed Journal)

Is Hereby Awarding this Certificate to

Chetan K. Wani

Student, Department of Information Technology Engineering, PES Modern COE, Pune, Maharashtra, India

In Recognition of the Publication of Manuscript Entitled

A Survey on Face Detection and Recognition Techniques

Published in

International Research Journal of Innovations in Engineering and Technology (IRJIET)

Volume 10, Issue 6, pp 193-200, June-2026

https://doi.org/10.47001/IRJIET/2026.106024

(2026) Impact Factor 8.7

IRJIET

Editor-In-Chief
IRJIET

Managing Editor
IRJIET

ISSN (online): 2581-3048

Mail us at: editor@irjiet.com / irjietjournal@gmail.com
Journal Website : www.irjiet.com

Manuscript ID : IRJIET106024
Date of Issue : June 22, 2026

CERTIFICATE OF PUBLICATION

IRJIET

INTERNATIONAL RESEARCH JOURNAL OF INNOVATIONS IN ENGINEERING AND TECHNOLOGY (IRJIET)

(International Open Access, Monthly, Peer Reviewed Journal)

Is Hereby Awarding this Certificate to

Shashikant K. More

Student, Department of Information Technology Engineering, PES Modern COE, Pune, Maharashtra, India

In Recognition of the Publication of Manuscript Entitled

A Survey on Face Detection and Recognition Techniques

Published in

International Research Journal of Innovations in Engineering and Technology (IRJIET)

Volume 10, Issue 6, pp 193-200, June-2026

https://doi.org/10.47001/IRJIET/2026.106024

(2026) Impact Factor 8.7

IRJIET

Editor-In-Chief IRJIET

Managing Editor IRJIET

ISSN (online): 2581-3048

Mail us at: editor@irjiet.com / irjietjournal@gmail.com

Journal Website : www.irjiet.com

Manuscript ID : IRJIET106024

Date of Issue : June 22, 2026

CERTIFICATE OF PUBLICATION

IRJIET

INTERNATIONAL RESEARCH JOURNAL OF INNOVATIONS IN ENGINEERING AND TECHNOLOGY (IRJIET)

(International Open Access, Monthly, Peer Reviewed Journal)

Is Hereby Awarding this Certificate to

Aaradhya S. Sonar

Student, Department of Information Technology Engineering, PES Modern COE, Pune, Maharashtra, India

In Recognition of the Publication of Manuscript Entitled

A Survey on Face Detection and Recognition Techniques

Published in

International Research Journal of Innovations in Engineering and Technology (IRJIET)

Volume 10, Issue 6, pp 193-200, June-2026

https://doi.org/10.47001/IRJIET/2026.106024

(2026) Impact Factor 8.7

IRJIET

Editor-In-Chief IRJIET

Managing Editor IRJIET

ISSN (online): 2581-3048

Mail us at: editor@irjiet.com / irjietjournal@gmail.com

Journal Website : www.irjiet.com

Manuscript ID : IRJIET106024

Date of Issue : June 22, 2026

Appendix B: Project Report Plagiarism Report

Dupli Checker

Date: 22-06-2026

Plagiarism Scan Report

2% Plagiarism

2% Exact Match

0% Partial Match

98% Unique

Words	1000
Characters	7934
Sentences	492
Paragraphs	363
Read Time	28 minute(s)
Speak Time	37 minute(s)

Content Checked For Plagiarism

CHAPTER 1

Introduction

1.1 Motivation

Every Every year, many people, including children, elderly individuals, and mentally challenged persons, are reported missing. Such incidents not only create emotional distress for families but also increase the workload on law enforcement agencies. Conventional methods used for locating missing individuals, such as poster circulation, newspaper advertisements, public announcements, and manual monitoring of CCTV footage, are often slow, inefficient, and highly dependent on human effort. In many cases, delays in identification reduce the chances of locating missing persons quickly.

Appendix C: Project Competition Participation/Achievement Certificates


INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

Mpulse

IDEA Lab

अभियन्त्रम्

A PROJECT COMPETITION

CERTIFICATE

OF PARTICIPATION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Aradhya Sonar

for participating in **"अभियंत्रम्"** A Project Competition organized by PES's Modern College of Engineering, Pune-05 under a grand technical event **"M-pulse (Feel the pulse)" 2026"** on Saturday, **11th** April 2026.

PROF. DR. K. R. JOSHI	PROF. DR. S. V. BHOSALE	PROF. DR. S. N. CHAPHEKAR
Principal, PESMCOE	Coordinator	Co-coordinator